

Job Market Analysis

Objective

The primary objective of this project is to analyze the job market for data science and analytics roles using 742 job postings. The project identifies hiring trends, salary patterns, industry demand, company hiring behavior, skill requirements, and educational influences on compensation. The business value is to support employers, recruiters, workforce planners, and job seekers with data-driven decision-making.

Dataset Overview

Dataset: Market. Total Records: 742. Total Companies: 343. Total Industries: 60. Major columns include Job_Title, Jobtitle_sim, Company_Name, Location, Industry, Avg_SalaryK, Lower_Salary, Upper_Salary, Degree, Python, SQL, Excel, Tableau, AWS, Spark, Hadoop, Keras, MongoDB, Revenue, Competitors, Headquarters and others. The dataset combines job market demand, salary information, organizational characteristics, and skill requirements.

Data Cleaning

Data preprocessing included assessment of missing values, handling of -1 placeholder values, validation of salary fields, removal of duplicate records where applicable, standardization of job titles, review of outliers, validation of categorical fields, and quality assurance checks. Placeholder values were treated as unavailable information and excluded or categorized appropriately during analysis.

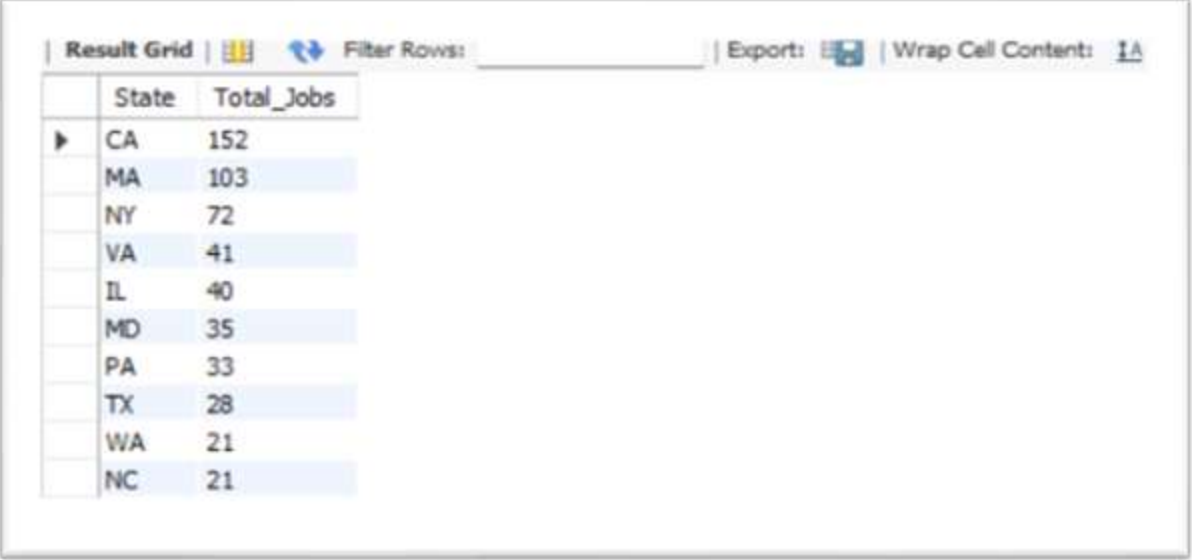
Analysis

Analysis focused on hiring locations, hiring industries, hiring companies, job roles, salary distribution, education impact, and skill demand. KPIs included Total Job Postings, Average Salary, Total Companies, and Total Industries. Aggregation, ranking, descriptive statistics, comparative analysis, and dashboard-based KPI monitoring were used throughout the project.

SQL Query Analysis

1. STATES WITH MOST NUMBER OF JOBS

```
SELECT RIGHT(Location, 2) AS State,  
COUNT(*) AS Total_Jobs  
FROM Market GROUP BY RIGHT(Location, 2)  
ORDER BY Total_Jobs DESC LIMIT 10;
```



The screenshot shows a SQL query result grid with a toolbar at the top containing 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. The table below displays the results of the query, listing the top 10 states by the number of jobs. The columns are 'State' and 'Total_Jobs'.

	State	Total_Jobs
▶	CA	152
	MA	103
	NY	72
	VA	41
	IL	40
	MD	35
	PA	33
	TX	28
	WA	21
	NC	21

2. AVERAGE MINIMAL AND MAXIMAL SALARIES IN DIFFERENT STATES

```
SELECT RIGHT(Location, 2),  
ROUND(AVG(Lower_Salary), 2) AS Avg_Min_Salary,  
ROUND(AVG(Upper_Salary), 2) AS Avg_Max_Salary  
FROM Market GROUP BY RIGHT(Location, 2)  
ORDER BY Avg_Max_Salary DESC LIMIT 10;
```

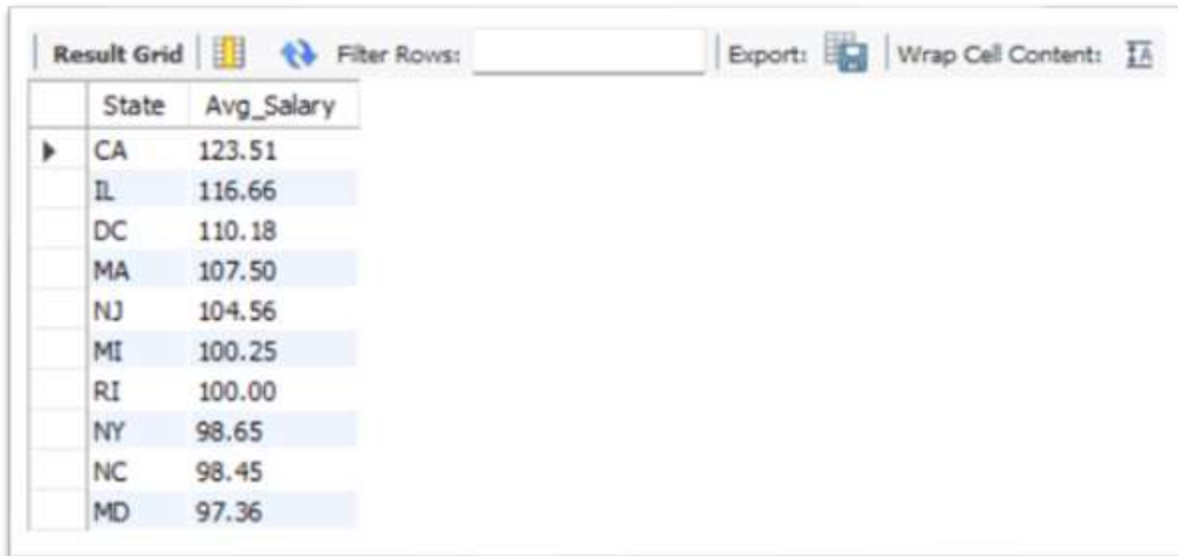


The screenshot shows a SQL query result grid with a toolbar at the top containing 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. The table below displays the results of the query, listing the top 10 states by average maximal salary. The columns are 'State', 'Avg_Min_Salary', and 'Avg_Max_Salary'.

	State	Avg_Min_Salary	Avg_Max_Salary
▶	CA	92.43	154.60
	IL	88.35	144.98
	MA	78.69	136.30
	NJ	76.82	132.29
	DC	88.64	131.73
	NC	69.05	127.86
	KY	66.33	127.67
	MD	68.31	126.40
	RI	74.00	126.00
	NY	72.13	125.18

3. AVERAGE SALARY IN DIFFERENT STATES

```
SELECT RIGHT(Location, 2),  
ROUND(AVG(Avg_SalaryK), 2) AS Avg_Salary  
FROM Market GROUP BY RIGHT(Location, 2)  
ORDER BY Avg_Salary DESC LIMIT 10;
```

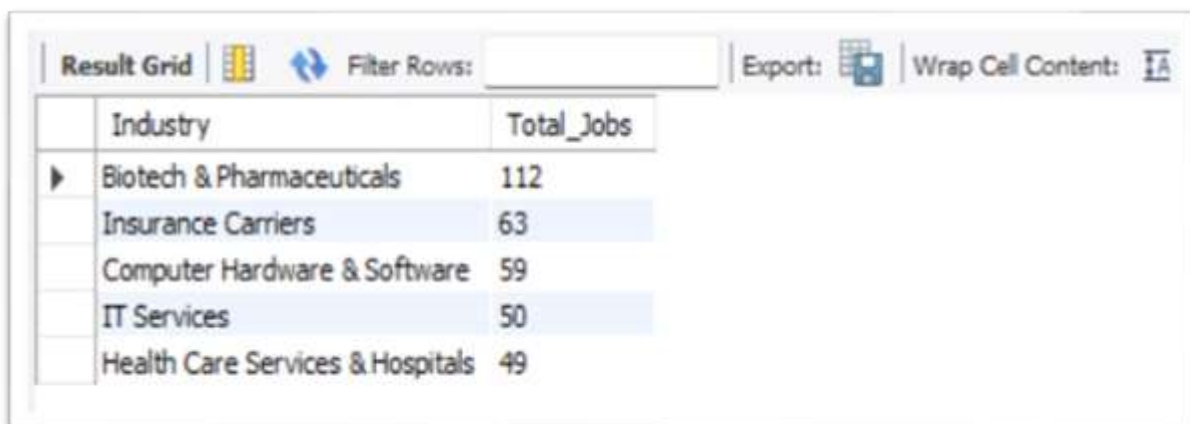


The screenshot shows a database query result grid with a toolbar at the top containing 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. The table has two columns: 'State' and 'Avg_Salary'. The data is sorted in descending order of average salary, with California (CA) having the highest average salary at 123.51 and Maryland (MD) having the lowest at 97.36 among the top 10 results.

State	Avg_Salary
CA	123.51
IL	116.66
DC	110.18
MA	107.50
NJ	104.56
MI	100.25
RI	100.00
NY	98.65
NC	98.45
MD	97.36

4. TOP 5 INDUSTRIES WITH MAXIMUM NUMBER OF DATA SCIENCE RELATED JOB POSTINGS

```
SELECT Industry,  
COUNT(*) AS Total_Jobs  
FROM Market WHERE Industry <> '-1' GROUP BY Industry  
ORDER BY Total_Jobs DESC LIMIT 5;
```



The screenshot shows a database query result grid with a toolbar at the top containing 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. The table has two columns: 'Industry' and 'Total_Jobs'. The data is sorted in descending order of the total number of jobs, with Biotech & Pharmaceuticals having the highest count at 112 and Health Care Services & Hospitals having the lowest at 49 among the top 5 results.

Industry	Total_Jobs
Biotech & Pharmaceuticals	112
Insurance Carriers	63
Computer Hardware & Software	59
IT Services	50
Health Care Services & Hospitals	49

5. COMPANIES WITH MAXIMUM NUMBER OF JOB OPENINGS

```
SELECT Company_Name, COUNT(*) AS Total_Openings
FROM Market GROUP BY Company_Name
ORDER BY Total_Openings DESC LIMIT 10;
```



The screenshot shows a database query result grid with the following data:

Company_Name	Total_Openings
MassMutual 3.6	14
Reynolds American 3.1	14
Takeda Pharmaceuticals 3.7	14
Software Engineering Institut...	11
Liberty Mutual Insurance 3.3	10
PNNL 3.8	10
AstraZeneca 3.9	9
MITRE 3.2	8
Rochester Regional Health 3.3	7
Fareportal 3.8	7

6. JOB TITLES WITH MOST NUMBER OF JOBS

```
SELECT Job_Title, COUNT(*) AS Total_Jobs
FROM Market GROUP BY Job_Title
ORDER BY Total_Jobs DESC LIMIT 10;
```



The screenshot shows a database query result grid with the following data:

Job_Title	Total_Jobs
Data Scientist	131
Data Engineer	53
Senior Data Scientist	34
Data Analyst	15
Senior Data Engineer	14
Senior Data Analyst	12
Lead Data Scientist	8
Sr. Data Engineer	6
Marketing Data Analyst	6
Principal Data Scientist	5

7. AVERAGE SALARY OF JOB TITLES WITH MOST NUMBER OF JOBS

```
SELECT Job_Title, COUNT(*) AS Total_Jobs,
ROUND(AVG(Avg_SalaryK),2) AS Avg_Salary
FROM Market GROUP BY Job_Title
ORDER BY Total_Jobs DESC LIMIT 10;
```

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	Job_Title	Total_Jobs	Avg_Salary			
▶	Data Scientist	131	106.18			
	Data Engineer	53	91.66			
	Senior Data Scientist	34	134.87			
	Data Analyst	15	66.30			
	Senior Data Engineer	14	121.93			
	Senior Data Analyst	12	83.42			
	Lead Data Scientist	8	161.25			
	Sr. Data Engineer	6	115.00			
	Marketing Data Analyst	6	48.67			
	Machine Learning Engineer	5	104.90			

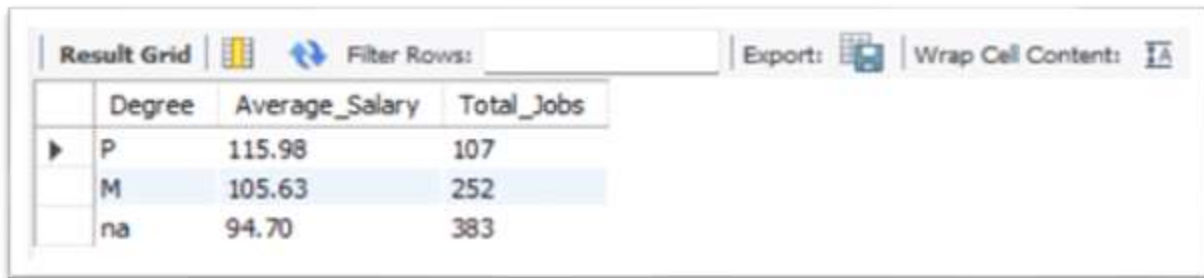
8. SKILLS REQUIRED BY COMPANIES FOR EACH JOB TITLE

```
SELECT job_title_sim,
SUM(Python) AS Python, SUM(sql_) AS SQL_Skill,
SUM(excel) AS Excel, SUM(tableau) AS Tableau,
SUM(aws) AS AWS, SUM(spark) AS Spark, SUM(hadoop) AS Hadoop,
SUM(sas) AS SAS, SUM(scikit) AS Scikit,
SUM(pytorch) AS PyTorch, SUM(keras) AS Keras,
SUM(mongo) AS MongoDB
FROM Market GROUP BY job_title_sim
ORDER BY job_title_sim;
```

Result Grid														Filter Rows:	Export:	Wrap Cell Content:
	job_title_sim	Python	SQL_Skill	Excel	Tableau	AWS	Spark	Hadoop	SAS	Scikit	PyTorch	Keras	MongoDB			
▶	analyst	31	75	77	40	10	6	3	11	0	0	0	3			
	data analytics	5	4	4	4	0	0	0	2	0	0	0	0			
	data engineer	77	87	54	11	59	67	50	1	0	0	0	13			
	data modeler	2	4	4	0	2	1	1	0	0	0	0	0			
	data scientist	240	176	155	76	79	84	60	50	47	33	29	19			
	Data scientist project manager	4	11	11	12	2	0	0	0	0	0	0	0			
	director	0	0	2	0	1	0	0	0	0	0	0	0			
	machine learning engineer	18	13	7	0	6	4	4	0	7	5	0	0			
	na	4	7	9	5	3	5	6	0	0	0	0	2			
	other scientist	11	3	65	0	14	0	0	2	0	1	0	0			

9. RELATION BETWEEN AVERAGE SALARY AND EDUCATION

```
SELECT Degree,  
ROUND(AVG(Avg_SalaryK),2) AS Average_Salary,  
COUNT(*) AS Total_Jobs  
FROM Market WHERE Degree <> '-1' GROUP BY Degree  
ORDER BY Average_Salary DESC;
```



The screenshot shows a SQL query result grid with the following data:

Degree	Average_Salary	Total_Jobs
P	115.98	107
M	105.63	252
na	94.70	383

Figure : | M = Master's degree | P = PhD's degree | na = None |

Insights and Business Interpretation

Highest Demand Role - Data Scientist is the most in-demand role with 131 openings, which is more than double the demand for Data Engineers (53 openings).

- This indicates that organizations are prioritizing Data Scientists to support data-driven decision making and AI initiatives.
- Job seekers can improve their career prospects by developing machine learning, statistics, and programming skills relevant to Data Scientist roles.

Python Dominates the Market - Python is the most frequently requested skill across Data Scientist, Data Engineer, and Analyst roles.

- Python has become the industry standard for data analysis, machine learning, and automation across multiple industries.
- Candidates with strong Python skills are likely to qualify for a wider range of analytics and data science positions.

SQL Is Non-Negotiable - SQL appears consistently across almost every job category and remains one of the most demanded skills in the dataset.

- SQL remains essential because organizations rely on databases for storing and managing business data.
- Mastering SQL alongside Python significantly improves employability for data-related roles.

Healthcare and Biotech Drive Hiring - Biotech & Pharmaceuticals generated 112 job postings, making it the largest hiring industry in the dataset.

- The increasing adoption of analytics in healthcare and biotechnology is creating strong demand for skilled professionals.
- Organizations in these industries require experts capable of extracting business and research insights from large datasets.

Education Impacts Salary - Jobs requiring PhD qualifications offer the highest average salary (115.98K), followed by Master's degree positions (105.63K).

- Higher educational qualifications often lead to higher-paying positions because they are associated with advanced technical knowledge and research expertise.
- Professionals aiming for senior or specialized roles may benefit from pursuing higher education or advanced certifications.

Seniority Increases Earnings - Lead Data Scientists earn an average salary of 161.25K, substantially higher than Data Analysts (66.30K).

- Senior-level professionals receive higher salaries due to greater experience, leadership responsibilities, and specialized expertise.
- Continuous learning and practical project experience help professionals progress toward higher-paying positions.

Location Matters More Than Most People Think - Bellevue, WA offers the highest average salary (184.50K), outperforming many traditional technology hubs.

- Salary levels and hiring demand vary significantly across locations because of differences in industry concentration and local economic conditions.
- Job seekers should evaluate both salary potential and regional hiring opportunities when planning their careers.

Cloud & Big Data Skills Are Valuable - AWS, Spark, and Hadoop frequently appear alongside Data Scientist and Data Engineer roles.

- The growing demand for AWS, Spark, and Hadoop reflects organizations' increasing reliance on cloud computing and big data technologies.
- Developing cloud and big data skills alongside Python and SQL can improve competitiveness for advanced analytics and data engineering roles.

Cross Analysis of Key Variables

Skills vs Salary

- Job roles requiring Python, SQL, AWS, and Spark generally offer higher average salaries.
- This suggests that professionals with multiple technical skills are rewarded with better compensation.

Industry Vs Hiring

- Healthcare & Biotechnology lead hiring because these industries increasingly rely on data-driven decision making.
- Companies in these sectors require professionals skilled in analytics and machine learning.

Location vs Opportunities

- Cities with the highest job postings also offer better career opportunities for data professionals.
- However, candidates should compare both salary and cost of living before relocating.

4. Education vs Salary

- Higher educational qualifications are associated with higher average salaries.
- Practical skills remain important, but advanced education provides an advantage for specialized positions.

5. Experience vs Salary

- Senior-level positions consistently receive higher salaries than entry-level roles.
- Continuous learning and industry experience are key factors for long-term salary growth.

Dashboard Analysis

DASHBOARD OVERVIEW

The dashboard provides an integrated view of job market trends, including hiring locations, industries, companies, salaries, education requirements, and skill demand. It serves as a decision-support tool for analysing workforce patterns and identifying key market opportunities.



Figure 1: Complete Job Market Analysis Dashboard summarizing major insights and hiring trends.

HIRING LOCATION ANALYSIS

This visualization presents the locations with the highest number of job postings in the dataset. Objective is to identify geographic regions with strong hiring demand for data science and analytics professionals. Understanding hiring concentration helps organizations target recruitment efforts and assists job seekers in identifying promising employment markets.

Business Interpretation: This suggests New York is a major analytics hub, making it an attractive location for professionals seeking greater career opportunities.

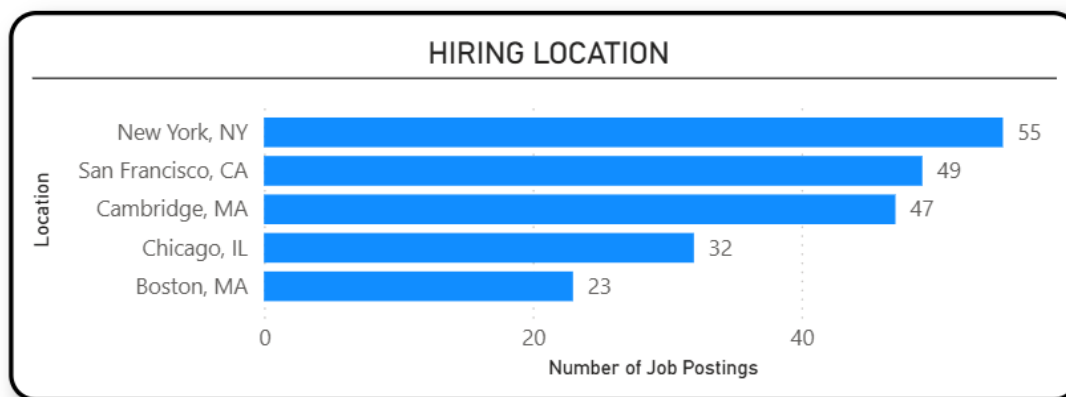


Figure 2: Hiring Location Analysis showing the locations with the highest number of job postings.

TOP HIRING INDUSTRIES

This chart highlights the industries generating the highest demand for data science and analytics professionals. The analysis helps identify sectors that are actively investing in data-driven decision-making and expanding their analytics workforce.

Business Interpretation: The dominance of Healthcare & Biotechnology indicates increasing investment in data-driven healthcare and research.

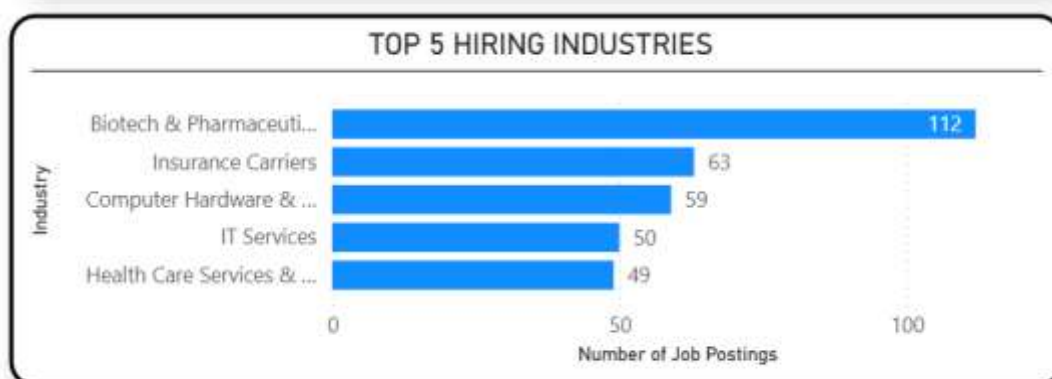


Figure 3: Top Hiring Industries showing industries with the maximum number of job postings.

TOP HIRING COMPANIES

This visualization identifies organizations with the largest number of job openings in the dataset. The chart helps evaluate company-level hiring activity and reveals employers that are aggressively recruiting data professionals.

Business Interpretation: Companies hiring repeatedly are likely expanding their analytics capabilities and require professionals with strong technical skills.

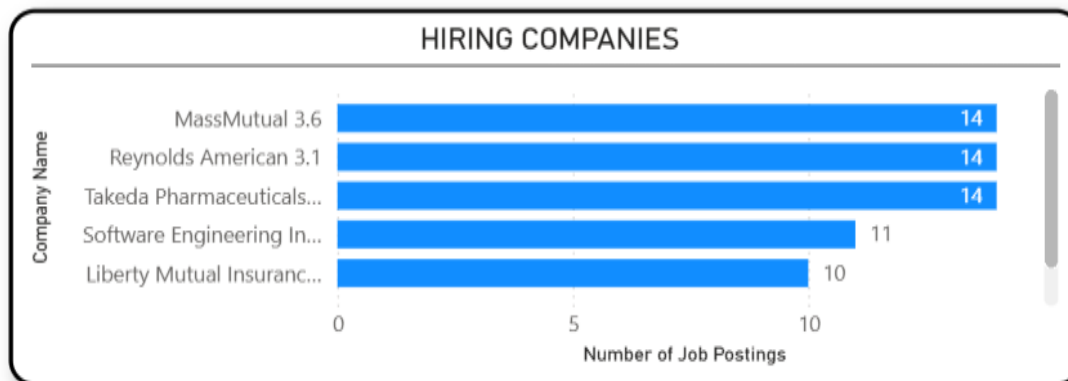


Figure 4: Hiring Companies showing organizations with the highest number of job postings.

JOB ROLE DISTRIBUTION

This chart illustrates the distribution of job roles across the dataset. The analysis helps determine which positions are most frequently advertised and provides insight into workforce demand across different data-related occupations.

Business Interpretation: The high demand for Data Scientists and Data Engineers highlights the industry's increasing focus on predictive analytics, machine learning, and large-scale data processing. Candidates preparing for these roles can expect broader career opportunities.

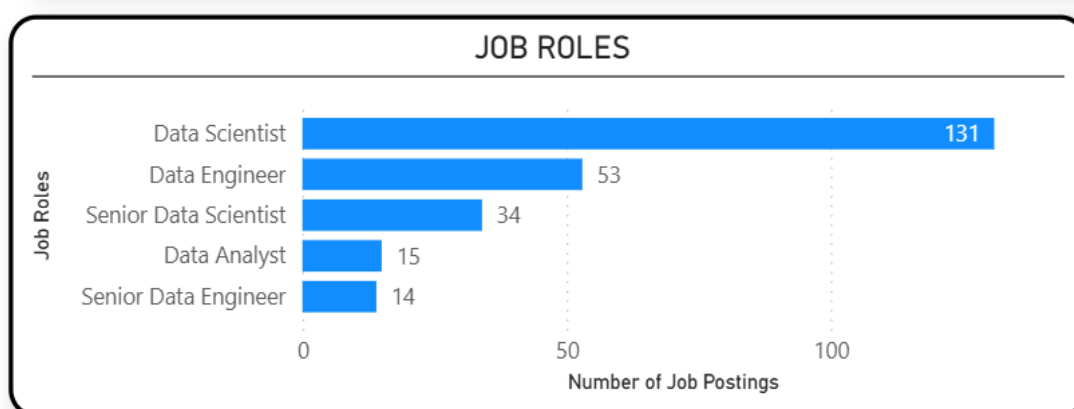


Figure 5: Job Role Distribution showing the most frequently advertised job roles.

AVERAGE SALARY BY JOB ROLE

This visualization compares average salaries across major job roles. The objective is to understand compensation trends and identify positions that offer the highest earning potential within the job market.

Business Interpretation: Specialized and senior analytical roles receive higher salaries because they require broader technical expertise and greater business responsibility.

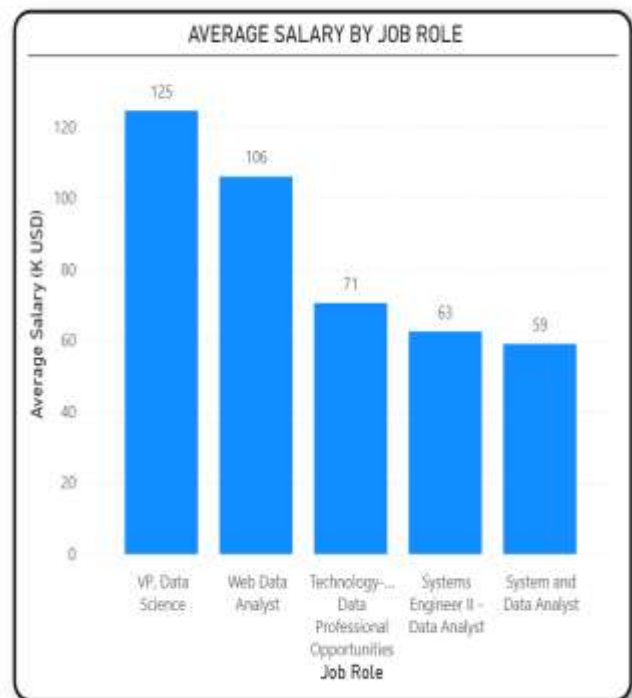


Figure 6: Average Salary by Job Role showing salary variations across different job positions.

AVERAGE SALARY BY EDUCATION LEVEL

This chart examines the relationship between educational qualifications and average salary. The analysis helps evaluate the impact of higher education on compensation and career opportunities.

Business Interpretation: Higher education provides a salary advantage, but employers also value practical technical skills and project experience.

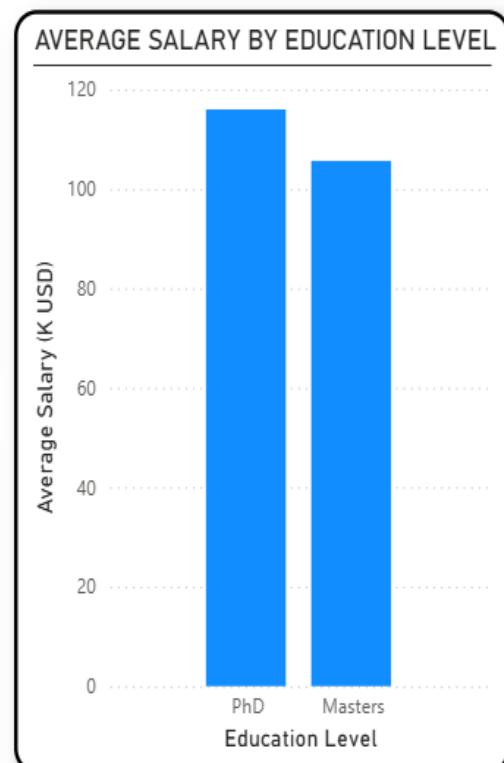


Figure 7: Average Salary by Education Level showing the relationship between education and compensation.

SKILLS REQUIRED BY JOB ROLE

This visualization presents the demand for key technical skills across different job roles. Skills such as Python, SQL, Excel, AWS, Tableau, and Spark are analysed to understand employer requirements and emerging market trends.

Business Interpretation: Python, SQL, AWS, and Spark frequently appear together, showing that employers increasingly expect a combination of programming, cloud, and big-data skills.

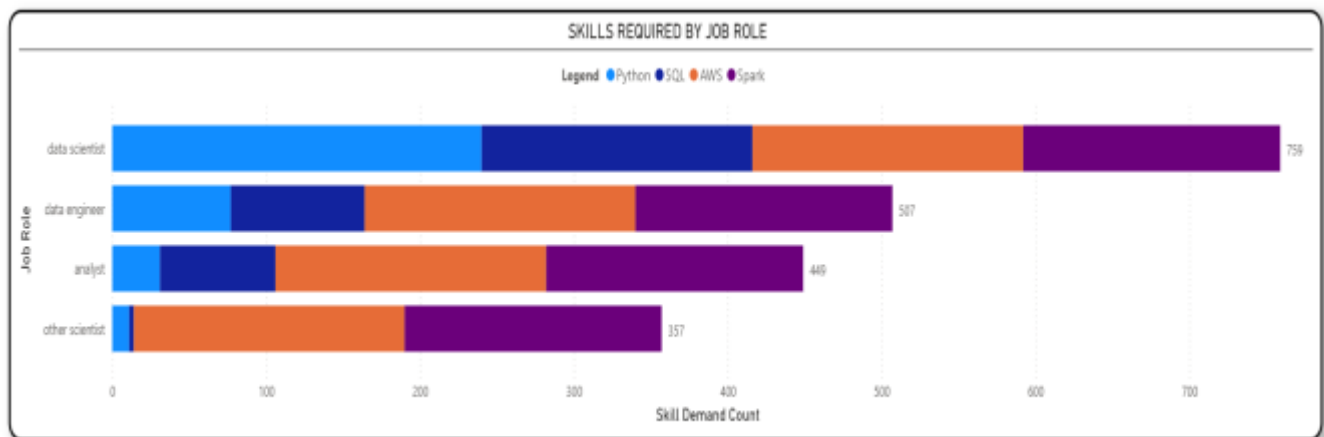


Figure 8: Skills Required by Job Role showing the distribution of technical skills across major job positions.

Overall Business Insights

- Data Scientist continues to be the most in-demand role, reflecting the rapid adoption of data-driven decision making across industries.
- Python and SQL remain the most essential technical skills, making them fundamental requirements for most data-related careers.
- Cloud technologies such as AWS and big data tools like Spark are becoming increasingly important as organizations process larger volumes of data.
- Healthcare & Biotechnology organizations are among the leading recruiters, demonstrating the growing role of analytics in healthcare innovation.
- Higher educational qualifications generally lead to better salary opportunities, although practical technical skills remain equally valuable.
- Major technology hubs provide significantly more employment opportunities for data professionals than smaller locations.
- Employers increasingly seek candidates with a combination of analytical, programming, cloud, and communication skills rather than expertise in a single technology.

Recommendations

For job Seekers

- Develop strong proficiency in Python and SQL, as they are consistently the most demanded technical skills.
- Learn cloud platforms such as AWS and big data technologies like Spark to remain competitive in the evolving job market.
- Build practical portfolio projects to demonstrate real-world analytical and technical capabilities.
- Focus on industries with strong hiring demand, particularly Healthcare & Biotechnology.
- Consider career opportunities in major technology hubs where hiring demand is significantly higher.

For Employers

- Invest in employee upskilling programs focusing on cloud computing and modern data technologies.
- Offer competitive compensation packages to attract highly skilled analytics professionals.
- Recruit candidates based on both practical project experience and academic qualifications.
- Expand hiring strategies beyond traditional talent pools to address increasing demand for skilled professionals.
- Encourage continuous learning and certification programs to improve workforce capabilities.

Conclusion

The project successfully converted raw job market data into actionable insights. Hiring demand, salary trends, industry opportunities, educational impact, and skill requirements were identified. The final dashboard and recommendations provide a strong decision-support framework for strategic workforce planning.

The analysis also identified relationships between technical skills, salaries, education level, industries, and hiring locations, enabling better career planning and workforce decision-making.